CCC-C-406b
April 25, 1967
SUPERSEDING
Fed. Spec. CCC-C406a
May 16, 1955

FEDERAL SPECIFICATION

CLOTH, AWNING, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1 SCOPE AND CLASSIFICATION

1 1 Scope This specification covers the requirements for cotton awning cloth

1 2 Classification

1 2 1 Types and classes The cotton cloth shall be of the following types and classes, as specified (see 6 2)

Type I—Awning cloth, yarn-dyed stripes
Class 1—8 40 ounces per square yard
Class 2—10.00 ounces per square yard
Class 3—12.00 ounces per square yard
Class 4—15 00 ounces per square yard

Type II—Awning cloth, painted stripes or painted solid colors

Type III—Awning cloth, piece-dyed

2 APPLICABLE DOCUMENTS

2 1 Specifications and standards The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein

Federal Specifications

CCC-D-950—Dyeing and Aftertreating Processes for Cotton Cloths
CCC-T-191—Textile Test Methods
PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics)

Federal Standard

Fed Std No 4—Glossary of Fabric Imperfections

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office, Washington, D C 20402

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D C , Atlanta, Chicago, Kansas City, Mo , Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies)

Military Standard

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

FSC 8305

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2 2 Other publications The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

American Trucking Associations, Inc

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc , 1616 P St , N W , Washington, D C 20036)

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained free of charge from the Federal Trade Commission, Washington, D C 20500)

Textile Color Card Association of the United States, Inc

Standard Color Card of America

(Application for copies should be addressed to the Textile Color Card Association of the United States, Inc , 200 Madison Avenue, New York, New York 10016)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W Jackson Blvd , Chicago, Illinois 20036)

Technical society and technical association specifications and standards are generally available for reference from libraries They are also distributed among technical groups and using Federal agencies

3 REQUIREMENTS

3 1 Standard sample When a standard sample is available the cloth shall match the sample with respect to all characteristics for which the standard sample is referenced (see 6 4)

3 1 1 Preproduction sample When required (see 6 2), before production is commenced, a three (3) yard full width sample of the finished cloth shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the production item

3.2 Material

3.2.1 Cotton The cotton shall be cleaned and carded, or combed The yarns shall be spun and twisted into ply yarns as specified in table I

TABLE I Physical requirements

Description	Width,[3] Inches	Weight, oz per sq yd [1]	Texture,[2]		Yarn Construction	Breaking Strength Lbs Minimum	
			Warp	Filling		Warp	Filling
Type I							
Class 1	31	8 5	61	28	2 ply Flat warp	150	70
Class 2	31	9 75	66	37	2 ply	175	95
Class 3	31	12 0	64	31	2 ply	210	100
Class 4	31	15.0	68	34	2 ply	240	120
Type II	32	10 0	—	—	—	150	150
Type III	31	10.0	54	42	2 ply	185	125

[1] A tolerance of minus 2½ percent or plus 5 percent will be permitted
[2] A minus tolerance of 3 yarns (total) and any plus tolerance will be permitted
[3] A minus tolerance of ¼ inch and any plus tolerance will be permitted

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3 2 2 Weave The weave shall be plain or modified plain

3 3 Physical requirements The cloth shall conform to the physical requirements specified in table I.

3 4 Color The color or color pattern shall be as specified (see 6 2) Unless otherwise specified the colors selected shall match the colors designated by cable numbers as cited in the Standard Color Card of America

3 4 1 Color matching The dyed cloth shall match the standard sample under natural (North Sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin

3 4 1 1 Dyeing The use of sulfur dye and dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited When a standard sample is available the dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur then the standard sample when tested as specified in 4 6 1 When no standard sample is available there shall be a minimum of staining of the test paper when tested as specified in 4 6 1

3 4 1 2 Colorfastness The finished treated cloth shall show fair colorfastness to dry crocking and accelerated weathering when tested as specified in 4 6

3 5 Length The length of the cloth shall be as specified (see 6 2)

3 6 Finish The finished treated cloth shall have a starch and protein content, including chloroform-soluble and water-soluble material of not more than 5 0 percent prior to application of the mildew resistant finish The mildew resistant finish shall comply with class A of CCC-D-950

3 7 Water resistance The water resistance of untreated cloth shall have a leakage of not more than 50 cubic centimeters at the height specified in table II when tested as specified in 4.6

TABLE II Water resistance

Weight of cloth	Hydrostatic height
Under 10 oz/sq yd	8 inches
10 to 15 oz/sq yd	10 inches

3 8 Breaking strength The breaking strength of the finished treated cloth after accelerated weathering shall be not less than 75 percent of the required strength specified in table I, when tested as specified in 4.6

3 9 Fiber identification Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act

3 10 Workmanship The awning cloth shall conform to the quality and grade of the product established by this specification Occurrence of defects shall not exceed the point level specified

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4 2 Lot All rolls of the same type and class offered to the Government under a specific contract, shall be considered a lot for purposes of inspection Examination of the end item shall be in accordance with the provisions of MIL-STD-105 and 4 4 through 4 5 3

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4.3 Preproduction sample inspection When required, the preproduction sample of the finished cloth submitted in accordance with 3 1 1 shall be visually examined for appearance, color, and finish The sample shall be tested for physical and chemical properties in accordance with methods specified in 4.6

4 4 Sampling

4 4 1 Sampling for examination A random sample of awning cloth shall be selected from each lot When the lot is large enough, the sample yardage shall be taken from 25 rolls The sample size shall be 750 linear yards, regardless of lot size The sample unit for examination shall be one linear yard An approximately equal number of yards shall be examined in each roll of the sample The lot shall be unacceptable if the points per hundred square yards exceeds 40 0 Point computation shall be as follows

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of material (inches)}} = \text{points per 100 square yards}$$

4.4 2 Sampling for tests A random sample of awning cloth shall be selected in accordance with table III The sample unit for testing shall be as follows one yard of the dyed cloth prior to mildew treatment, three continuous yards full width of the finished cloth The lot shall be unacceptable if one or more units fail to meet any requirements specified The lot size shall be expressed in units of one linear yard

TABLE III Sample size

Lot size in yards	Sample size (no of units)
800 or less	2
801 to 22,000	3
22,001 and over	5

4 5 Examination

4 5 1 Visual examination The samples selected in accordance with 4 4 1 shall be examined to verify compliance with this specification The required yardage of each roll shall be examined on the face side and all defects defined in section 1 of Fed Std No 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as follows

For defects 3 inches or less in any dimension — one point

For defects exceeding 3 inches but not exceeding 6 inches in any dimension — two points

For defects exceeding 6 inches but not exceeding 9 inches in any dimension — three points

For defects exceeding 9 inches in any dimension — four points

Slubs and knots which exceed the maximum limitation shown for each (see figure 1) shall also be scored

The following defects when present shall be scored four points for each yard in which they occur

Baggy, ridgy, or wavy cloth

Color not uniform, off shade, (not within established tolerance) shaded end to end, side to side, or side to center

Poor dye penetration, mottled, streaky or cloudy

Overall application of compound not uniform

Objectionable odor

Overall uncleanness

Clearly visible crystallization of mildew inhibitors

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Poorly constructed—not firmly or tightly woven

Bias or bowed filling

Tackiness (sticky to the touch)

Width less than specified

4 5 2 Examination for length

4 5 2 1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length Any length found to be less than the minimum specified or more than two yards under the length marked on the ticket shall be considered a defect with respect to length The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length

4 5 2 2 Total yardage in sample The lot shall be unacceptable if the total of the actual length of rolls in the sample is less than the total of the lengths marked on the ticket

4 5 3 Examination for compliance with Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for compliance with the Rules and Regulations of the Textile Fiber Products Identification Act Each roll not labeled in accordance with this act shall be a defect The lot shall be unacceptable if two or more of these defects occur

4 6 Testing of the end item The method of testing specified in CCC-T-191, whenever applicable, and as listed in table IV, shall be followed The physical and chemical values specified in section 3 (except where otherwise indicated) apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods All physical tests shall be performed under standard conditions as defined in CCC-T-191 When the data in the "number of determinations per individual sample unit" and "results reported as" columns are not specified in table IV, they shall be as required by the referenced test method

TABLE IV Test method

Characteristic	Reqt Para	Test Method	No of Dets	Results Reported as
Base cloth				
Weight	Table I	5041	—	—
Texture	Table I	5050	—	—
Breaking strength	Table I	5100	–	—
Material Identification	3.2 1	(1)	—	—
Yarn construction	Table I	(1)	—	—
Water resistance	3 7	5516[3]	—	—
Finished cloth				
Size content	3 6	2611	—	—
Sulfur dye	3 4 1.1	4 4 2	1	Pass or fail
Colorfastness to				
Dry crocking	3 4 1 2	5650	—	—
Accelerated weathering	3.4 1 2	5670 [2]	—	—
Breaking strength	3 8	5100	—	—

[1] Certificate of compliance is required for these characteristics
[2] With filters
[3] Except that the specimen shall be immersed in water to a depth of 2 inches for 1 hour and shall be tested while still wet

1 6 1 Test for sulfur content.

1 6 1 1 Solution (acid stannous chloride) Dissolve 100 grams of stannous chloride crystals in 100 ml of hydrochloric acid (35 percent concentration) and add 50 ml of distilled water

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4 6 1 2 Procedure One to 1 5 grams of the finished cloth, cut into small pieces shall be used for the test Place the sample in a test tube of suitable size and cover with stannous chloride solution Place a filter paper which has been moistened with one drop of 5 0 percent lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 ml of distilled water, add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the tube (punch needle hole in paper) Heat the test tube slowly over a low flame until the solution is boiling Solution shall not be boiled for more than 10 to 15 seconds, as prolonged boiling will reduce the sensitivity of the test A brown to black stain appearing on the paper indicates the presence of sulfur

4 6 1.3 Evaluation The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory When no standard sample is available there shall be a minimum of staining on the paper

4 7 Inspection of preparation for delivery The packaging, packing, and marking of the cloth shall be inspected to determine compliance with the applicable requirements of section 5 of this specification

5 PREPARATION FOR DELIVERY

5.1 Packaging Packaging shall be level A or C as specified (see 6 2)

5 1.1 Level A The cloth shall be packaged in accordance with the applicable level A requirements of PPP-P-51

5.1.2 Level C The cloth shall be packaged in accordance with the supplier's standard practice

5.2 Packing Packing shall be level A, B, or C as specified (see 6 2)

5 2 1 Level A The cloth shall be packed in accordance with the applicable level A requirements of PPP-P-51

5 2 2 Level B The cloth shall be packed in accordance with the applicable level B requirements of PPP-P-51.

5.2 3 Level C The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates Containers and packing shall comply with Uniform Freight Classification Rules or with National Freight Classification Rules

5 3 Marking (see 6 2) In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-51

6 NOTES

6 1 Intended use The awning cloth is intended for use as window awnings, canopies, patio awnings, poolside accessories, fences, and windbreakers

6.2 Ordering data Purchasers should select the preferred options permitted herein and include the following information in procurement documents

(a) Title, number, and date of this specification
(b) Type and class of cloth required (see 1 2)
(c) Whether a preproduction sample is required (see 3 1 1)
(d) Color of cloth required (see 3 4)
(e) Length of cloth required (see 3 5)
(f) Level of packaging and level of packing required (see 5 1 and 5 2)
(g) Marking required (see 5 3)

6 2 1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives as indicated in PPP-P-51 may be exercised in the preparation for the delivery of cloth

6.3 When level B packaging requirements are specified by civil agency procurement officers, the requirements in 5.1.1 shall apply

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6 4 Standard sample For access to standard sample, address the procuring office issuing the invitation for bids.

MILITARY INTEREST

Custodians·
Navy—YD

Review·
Navy—YD, SA

Code "C"

Preparing activity
Navy—YD

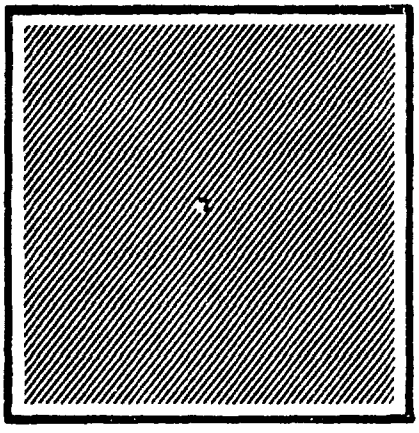
Civilian agencies
Interest
GSA—FSS

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LIMITS OF ACCEPTABILITY

Knot or Lump



Yarn Thickening (Slub)

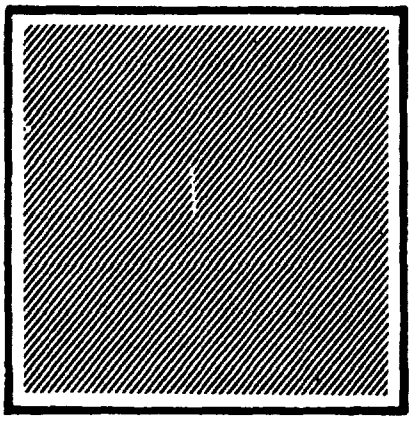


Figure 1

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents See section 2 of this specification to obtain extra copies and other documents referenced herein Price 5 cents each

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

CCC-C-406b: Cloth, Awning, Cotton

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 IA 0101-807-2000 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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